

Air Pollution Levels and Health Outcomes on Hospital Emergency Department Visits for HCA Healthcare Resource Allocation and Facility Preparation

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Business Problem



HCA Healthcare Air Quality Impact Analysis

Background

- HCA Healthcare is nation's leading healthcare system, operating 219 hospitals with over 41,600 beds
- Generates highest net patient revenue in the United States
- Positioned as critical player in addressing emerging public health challenges
- Escalating concern over health impacts created by air pollution and urgent pressures on healthcare systems



Business Problem and Case

Business Question

- How can HCA Healthcare leverage predictive analytics to optimize resource allocation and facility preparedness across its 219 hospitals in response to air quality-driven visits patterns?

Business Case

- Challenge spans multiple operational areas including patient care strategy, revenue management, direct costs, and operational efficiency across diverse geographic regions with varying air quality concerns



Business Problem and Case

Project Approach

- Implementation of advanced analytics and machine learning methodologies to:
 - Examine correlations between air quality metrics and public health outcomes
 - Analyze emergency room visit patterns
 - Develop predictive models for resource planning
 - Generate actionable insights for strategic decision-making

Expected Outcomes

- Development of data-driven strategies to:
 - Enhance preparedness for air quality-related health events
 - Optimize resource allocation across HCA's hospital network
 - Improve operational efficiency and patient care delivery
 - Support strategic planning and decision-making processes



02

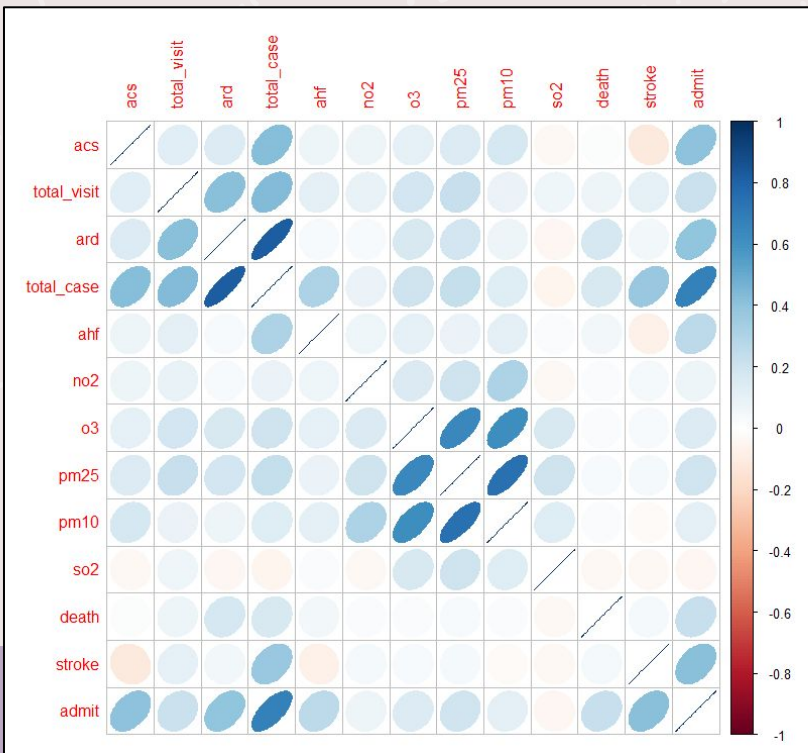
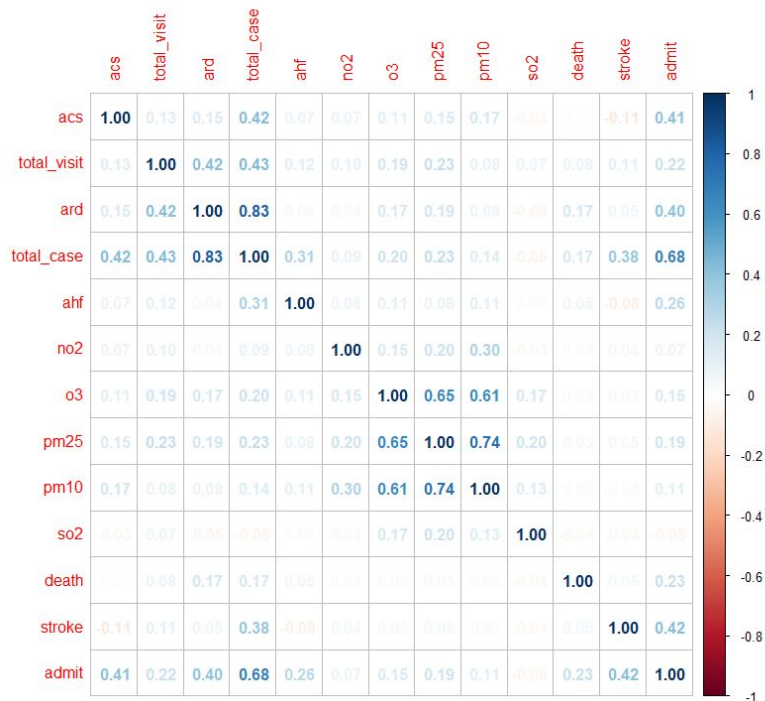
Description of Dataset



Dataset

Variable	Range	Type	Unit	Description	R Class
Time Variables					
day	(1-365)	Discrete numeric	Day sequence number	Sequential day counter throughout study period	integer
date	m/d/y	Temporal	Calendar date	Daily date from April 2018 to March 2019	character
Disease-Specific Cases					
ard	(0-16)	Discrete numeric	Number of cases per day	Daily count of acute respiratory disease cases	integer
acs	(0-8)	Discrete numeric	Number of cases per day	Daily count of acute coronary syndrome cases	integer
ahf	(0-5)	Discrete numeric	Number of cases per day	Daily count of acute heart failure cases	integer
stroke	(0-8)	Discrete numeric	Number of cases per day	Daily count of stroke cases	integer
Hospital Metrics					
total_case	(2-23)	Discrete numeric	Number of cases per day	Total daily cases of the four specific conditions combined	integer
total_visit	(27-92)	Discrete numeric	Number of visits per day	Total daily Emergency Department visits	integer
admit	(0-13)	Discrete numeric	Number of admissions per day	Total daily hospital admissions	integer
death	(0-3)	Discrete numeric	Number of deaths per day	Total daily deaths	integer
Air Quality Metrics					
pm25	(39-282)	Continuous numeric	µg/m³	Daily PM2.5 concentration	integer
pm10	(9-176)	Continuous numeric	µg/m³	Daily PM10 concentration	integer
o3	(2-53)	Continuous numeric	ppb	Daily ozone concentration	integer
no2	(3-45)	Continuous numeric	ppb	Daily nitrogen dioxide concentration	integer
so2	(0-27)	Continuous numeric	ppb	Daily sulfur dioxide concentration	integer

Correlations



Visual Plot Inspections

Residuals vs Fitted Plot

- Relatively random scatter indicating decent linear fit
- Some outliers

Q-Q Plot

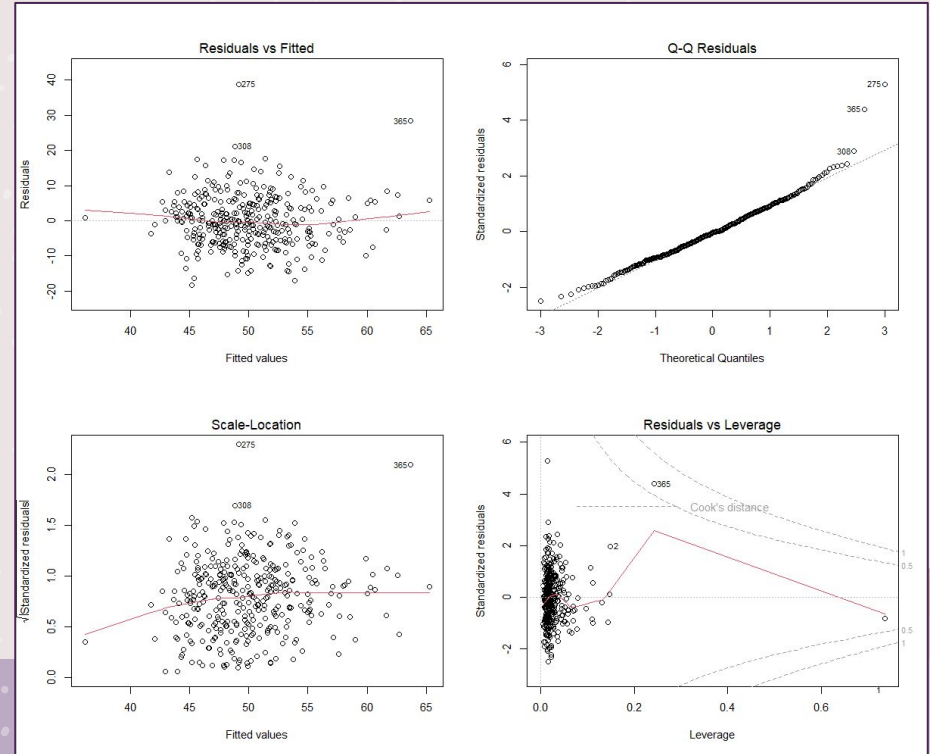
- Shows approximate normality along line
- Some concerning deviations at both tails

Scale-Location Plot

- Shows some heteroscedasticity
- Some visible clustering patterns

Residuals vs Leverage Plot

- Most points have low leverage, showing good model stability



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Explanation of Model



Model Selection and Alternatives

- Comprehensive testing of regression assumptions
 - VIF scores for multicollinearity
 - Breusch-Pagan tests for heteroscedasticity
 - Durbin-Watson tests for serial correlation
 - Visual inspection of plots
- Stepwise selection
- ANOVA tests for model comparisons



Pre-processing and OLS Model

- Data clean-up - missing values for o3 (used mean)
- Limited predictive power
 - R-squared values ~22-23%
 - Few significant predictors
- Tests
 - Variance Inflation Factor Test
 - Breusch-Pagan Test
 - Durbin-Watson Test

```
> vif(OLSm1)
      ard      acs      ahf      stroke      pm25      pm10      o3      no2      so2
1.080920 1.069871 1.024472 1.037529 2.707215 2.620607 1.863985 1.115462 1.066192
> bptest(OLSm1)

studentized Breusch-Pagan test

data:  OLSm1
BP = 14.389, df = 9, p-value = 0.1091

> durbinWatsonTest(OLSm1)
lag Autocorrelation D-W Statistic p-value
1      0.07782565      1.802544      0.048
Alternative hypothesis: rho != 0
```

```
Call:
lm(formula = total_visit ~ ard + acs + ahf + stroke + pm25 +
    pm10 + o3 + no2 + so2, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-18.206  -5.078  -0.256   4.664  38.850

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 38.43336    1.49160  25.766  < 2e-16 ***
ard           1.13318    0.14751   7.682 1.54e-13 ***
acs           0.48297    0.32949   1.466  0.14358
ahf           0.78795    0.37704   2.090  0.03734 *
stroke        0.50046    0.27575   1.815  0.07038 .
pm25          0.04374    0.01585   2.759  0.00610 **
pm10         -0.08103    0.02961  -2.737  0.00652 **
o3            0.08234    0.07347   1.121  0.26320
no2           0.09264    0.06460   1.434  0.15241
so2           0.31635    0.24369   1.298  0.19508
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.394 on 355 degrees of freedom
Multiple R-squared:  0.2399,    Adjusted R-squared:  0.2206
F-statistic: 12.45 on 9 and 355 DF, p-value: < 2.2e-16
```

Transformations

Multivariate Time Series Model

- Created lagged variables for both health conditions and environmental factors up to 3 days
- Removed NA rows
- Disappointing results
 - Did not improve OLS model
 - Decreased R-squared values
 - Loss of significant predictors

```
Call:
lm(formula = total_case ~ pm25 + pm10 + o3 + no2 + so2 + pm25.L1 +
    pm25.L2 + pm25.L3 + pm10.L1 + pm10.L2 + pm10.L3 + o3.L1 +
    o3.L2 + o3.L3 + no2.L1 + no2.L2 + no2.L3 + so2.L1 + so2.L2 +
    so2.L3 + total_case.L1 + total_case.L2 + total_case.L3, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.3783	-2.4902	-0.2858	1.8685	11.9839

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	6.9316448	0.9801485	7.072	9.17e-12	***
pm25	0.0173587	0.0188822	0.919	0.3586	
pm10	-0.0219501	0.0144334	-1.521	0.1293	
o3	0.0923053	0.0432971	2.132	0.0338	*
no2	0.0306549	0.0692178	0.443	0.6581	
so2	-0.3692571	0.3498287	-1.056	0.2920	
pm25.L1	0.0322161	0.0205642	1.567	0.1182	
pm25.L2	0.0034273	0.0202922	0.169	0.8660	
pm25.L3	-0.0014308	0.0110332	-0.130	0.8969	
pm10.L1	0.0077221	0.0265476	0.291	0.7713	
pm10.L2	-0.0525402	0.0275273	-1.909	0.0572	.
pm10.L3	-0.0026972	0.0269894	-0.100	0.9205	
o3.L1	-0.0323019	0.0471137	-0.686	0.4934	
o3.L2	-0.0410071	0.0471990	-0.869	0.3856	
o3.L3	-0.0365140	0.0447020	-0.817	0.4146	
no2.L1	-0.0389821	0.1042253	-0.374	0.7086	
no2.L2	-0.0241421	0.1053523	-0.229	0.8189	
no2.L3	0.0823690	0.0729842	1.129	0.2599	
so2.L1	0.0293417	0.3561298	0.082	0.9344	
so2.L2	0.3324222	0.3232878	1.028	0.3046	
so2.L3	-0.0219099	0.1632055	-0.134	0.8933	
total_case.L1	0.0576917	0.0553789	1.042	0.2983	
total_case.L2	0.0657754	0.0553548	1.188	0.2356	
total_case.L3	0.0003828	0.0556007	0.007	0.9945	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.541 on 330 degrees of freedom
Multiple R-squared: 0.1117, Adjusted R-squared: 0.04979
F-statistic: 1.804 on 23 and 330 DF, p-value: 0.01418

Transformations

Stepwise Selection and ANOVA test

- Stepwise Selection to simplify model and identify significant predictors
 - Improved efficiency but still offered limited explanatory power
 - R-squared increased slightly from 0.049 to 0.072
 - Identified key predictors: current-day effects of PM2.5, PM10, and O3 and lagged effects of PM2.5 (one-day), PM10 (two-day), O3 (three-day), and NO2 (three-day)
- ANOVA test suggested Stepwise better model
 - Achieved similar performance with fewer predictors, making it simpler and more interpretable
 - Helped identify most significant predictors but suggested need for more robust approach to capture complex relationships in dataset

```
> anova(Lagmodel1, Stepmodel1)
Analysis of Variance Table

Model 1: total_case ~ pm25 + pm10 + o3 + no2 + so2 + pm25.L1 + pm25.L2 +
  pm25.L3 + pm10.L1 + pm10.L2 + pm10.L3 + o3.L1 + o3.L2 + o3.L3 +
  no2.L1 + no2.L2 + no2.L3 + so2.L1 + so2.L2 + so2.L3 + total_case.L1 +
  total_case.L2 + total_case.L3
Model 2: total_case ~ pm25 + pm10 + o3 + pm25.L1 + pm10.L2 + o3.L3 + no2.L3
Res.Df    RSS   Df Sum of Sq    F Pr(>F)
1      338 4205.6
2      354 4298.1 -16   -92.511  0.4647 0.9622
>
```


Transformations

Simplified Log-transformed Model with interactions (Final model)

- Centered predictor variables to reduce multicollinearity and improve interpretation of interaction effects
- Log-transformed the total case count to address potential non-linearity
- Results
 - Improved R-squared = 0.9439
 - Highly significant p-value < 0.001
 - More significant predictors
 - VIF scores < 10 (no multicollinearity)
 - BP test p-value = 0.344 (homoscedastic)
 - DW test = 1.949 (no serial correlation)
- Captured complex interaction effects between pollutants and health conditions

Call:

```
lm(formula = log_total_case ~ pm25_c + o3_c + pm10_c + no2_c +  
  so2_c + ard_c + acs_c + ahf_c + stroke_c + pm25_c:ard_c +  
  pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c + log_total_case.L1,  
  data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.60493	-0.02857	0.02733	0.05434	0.11990

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.25056597	0.02929135	76.834	< 2e-16 ***
pm25_c	0.00035111	0.00019647	1.787	0.074798 .
o3_c	-0.00062660	0.00091670	-0.684	0.494725
pm10_c	0.00009012	0.00036728	0.245	0.806311
no2_c	-0.00013774	0.00079111	-0.174	0.861880
so2_c	0.00445839	0.00829669	0.537	0.591358
ard_c	0.10143608	0.00184141	55.086	< 2e-16 ***
acs_c	0.09932985	0.00405151	24.517	< 2e-16 ***
ahf_c	0.10095160	0.00466409	21.644	< 2e-16 ***
stroke_c	0.10884061	0.00336797	32.316	< 2e-16 ***
log_total_case.L1	-0.01758226	0.01308719	-1.343	0.180000
pm25_c:ard_c	-0.00016469	0.00006308	-2.611	0.009428 **
pm25_c:ahf_c	-0.00040309	0.00011655	-3.458	0.000611 ***
pm25_c:stroke_c	-0.00031829	0.00008323	-3.824	0.000155 ***
o3_c:ard_c	-0.00121841	0.00035323	-3.449	0.000631 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08977 on 346 degrees of freedom
(1 observation deleted due to missingness)

Multiple R-squared: 0.9461, Adjusted R-squared: 0.9439

F-statistic: 433.8 on 14 and 346 DF, p-value: < 2.2e-16

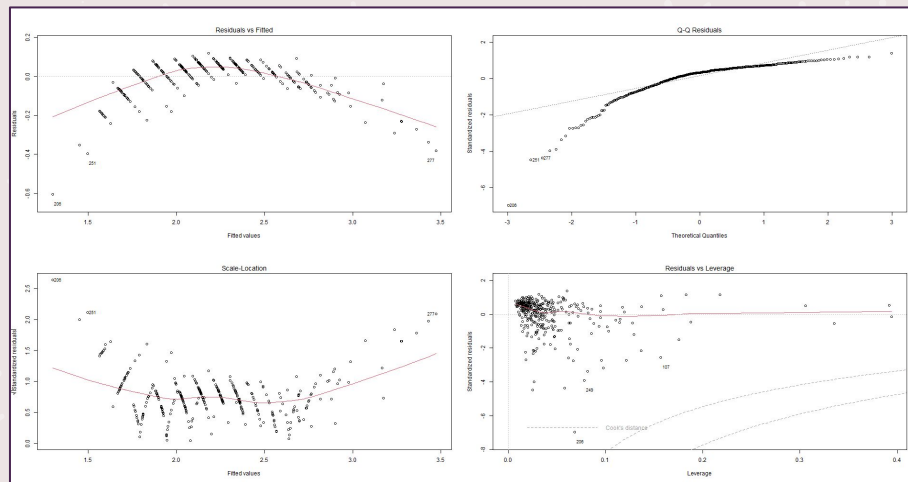
Model Specifications

```
> bptest(Finalmodel)

studentized Breusch-Pagan test

data: Finalmodel
BP = 15.511, df = 14, p-value = 0.3441

> durbinWatsonTest(Finalmodel)
lag Autocorrelation D-W Statistic p-value
1 0.02472321 1.949329 0.492
Alternative hypothesis: rho != 0
```



pm25_c	o3_c	pm10_c	no2_c	so2_c	ard_c	acs_c	ahf_c	stroke_c	log_total_case.L1
2.770649	1.936976	2.724988	1.132246	1.139419	1.133999	1.081324	1.036443	1.044411	1.100446
pm25_c:ard_c	pm25_c:ahf_c	pm25_c:stroke_c	o3_c:ard_c						
1.880845	1.112242	1.086990	1.912393						

Model Specifications

Evaluated models based on statistical performance, complexity, and practical interpretability



Best overall

High R-squared values
Significant predictors



Meets key regression assumptions

Passed tests and/or
visual inspection of
plots



Maintains significant health condition interactions



Parsimonious

Cross-Validation Test

Results provide strong evidence that model will be reliable for predicting hospital visits based on air pollution and health outcome variables

- RMSE = 0.0992
 - Average prediction error in log units
 - Approximate 9.92% error in predicting total cases
 - Small error given scale of data
- Cross-validated R-squared = 0.9374
 - Model explains approximately 93.74% of the variance in out-of-sample predictions
 - Close to Final Model's Adjusted R-squared of 0.9439
 - Suggests model is not overfitting
- MAE = 0.0671
 - Measured in log units
 - Indicates that prediction errors are approximately 6.71%
 - Because MAE is lower than the RMSE, shows that predictions are generally consistent with few extreme errors

```
Linear Regression
361 samples
10 predictor

No pre-processing
Resampling: Bootstrapped (25 reps)
Summary of sample sizes: 361, 361, 361, 361, 361, 361, ...
Resampling results:

RMSE      Rsquared    MAE
0.09928906 0.9374639 0.06716341

Tuning parameter 'intercept' was held constant at a value of TRUE
```

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Results



Most Relevant Results

Final model demonstrates robust predictive power:

- Multiple R-squared = 0.9461
- Adjusted R-squared = 0.9439
- P-value = < 0.001 ***
- Residual standard error = 0.0897
 - On average, predicted values deviate from actual values by approximately 8.97%



Most Relevant Results

Health conditions emerged as strongest predictors of total cases, with each condition showing substantial impact:

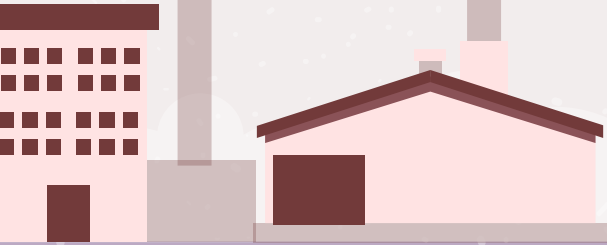
- ARD: 10.09% increase per unit (p-value < 0.001 ***)
- ACS: 9.93% increase per unit (p-value < 0.001 ***)
- AHF: 10.09% increase per unit (p-value < 0.001 ***)
- Stroke: 10.88% increase per unit (p-value < 0.001 ***)

Model identified significant negative interactions between pollutants and health conditions:

- PM2.5 × ARD: $\beta = -0.00016$ (p-value < 0.01 **)
- PM2.5 × AHF: $\beta = -0.00040$ (p-value < 0.001 ***)
- PM2.5 × Stroke: $\beta = -0.00031$ (p-value < 0.001 ***)
- O3 × ARD: $\beta = -0.00121$ (p-value < 0.001 ***)

05

Conclusion



Final Conclusion

- Health conditions are primary drivers of ED visits
- Air pollution acts as significant moderator
- Different health conditions show varying sensitivities to air pollution
- Resource allocation recommendations:
 - Prioritize preparation for specific health conditions
 - Consider air quality as moderating factor
- Air pollution and health outcomes relationship:
 - Exhibits complex, non-linear patterns
 - Different pollutants impact specific conditions differently
- Practical application:
 - Air quality monitoring can serve as early warning system
 - Help predict potential increases in hospital visits

06

Challenges



Challenges Encountered



Model Selection

- DIFFICULT PROCESS!
- Consistently low and decreasing R-squared values before centering data
- Attempted most transformation models (Ridge, Lasso, etc.)
- Multivariate Time Series in one iteration resulted in perfect multicollinearity issue (Adjusted R-squared = 1)



Variables and Interactions

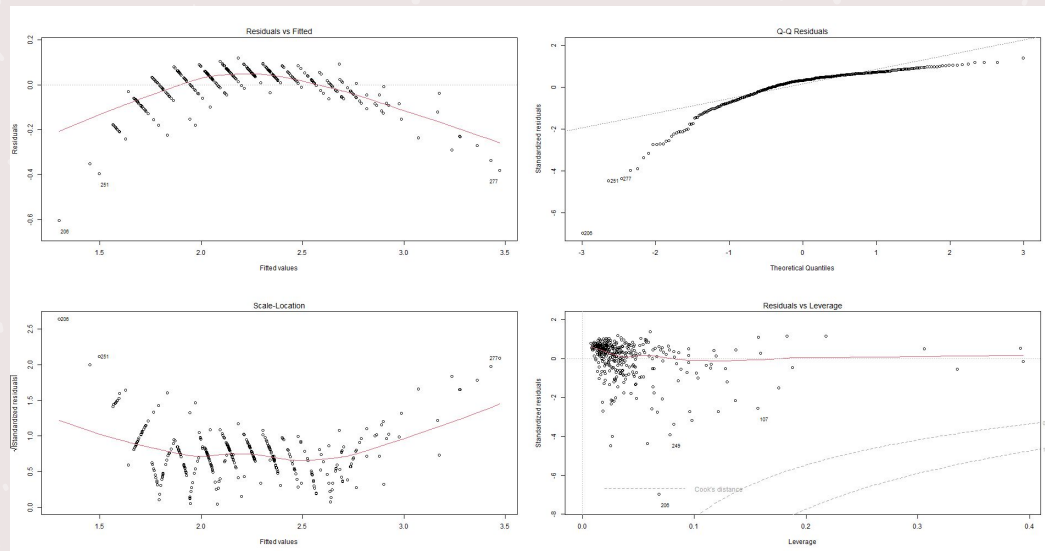
- Climate science and health expertise – reminder that we are not trying to save the world!
- Finding right balance between transformations and interactions
- Logistic regression model vs. Classification model (creating pollution levels for alert system)
- Attempted several models with groups
 - Seasons (cool, hot, and rainy)
 - Weighted pollution index (PM2.5 60%, PM10 20%, O3 5%, NO2 5%, and SO2 5%)

Challenges Encountered



Correcting data

- Little to no improvements between models until centering data for interactions log model
- Some curvilinear relationships in final model



Thank you!

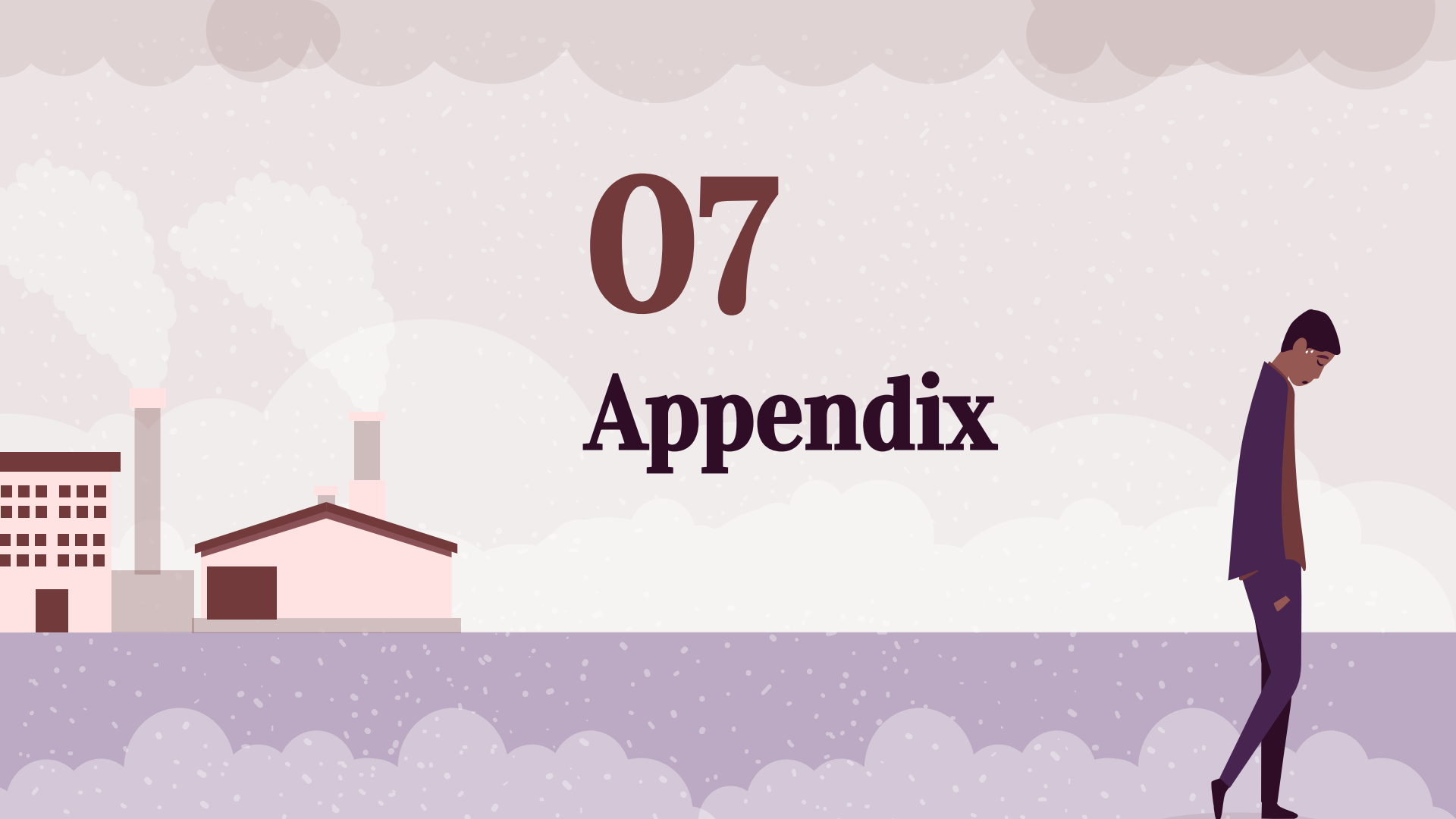
Questions?

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07

Appendix



Appendix A: Summary of dataset

```
> summary(data)
```

day	date	ard	acs	ahf	stroke	total_case	total_visit	admit	death	pm25
Min. : 1	Length:365	Min. : 0.000	Min. :0.000	Min. :0	Min. :0.000	Min. : 2.000	Min. :27.00	Min. : 0.000	Min. :0.0000	Min. : 39.00
1st Qu.: 92	Class :character	1st Qu. : 3.000	1st Qu.:0.000	1st Qu.:0	1st Qu.:1.000	1st Qu.: 7.000	1st Qu.:44.00	1st Qu.: 3.000	1st Qu.:0.0000	1st Qu.: 63.00
Median :183	Mode :character	Median : 5.000	Median :1.000	Median :1	Median :2.000	Median : 9.000	Median :49.00	Median : 4.000	Median :0.0000	Median : 75.00
Mean :183		Mean : 5.405	Mean :1.288	Mean :1	Mean :2.005	Mean : 9.699	Mean :49.88	Mean : 4.559	Mean :0.4192	Mean : 88.96
3rd Qu.:274		3rd Qu. : 7.000	3rd Qu.:2.000	3rd Qu.:2	3rd Qu.:3.000	3rd Qu.:11.000	3rd Qu.:55.00	3rd Qu.: 6.000	3rd Qu.:1.0000	3rd Qu.:102.00
Max. :365		Max. :16.000	Max. :8.000	Max. :5	Max. :8.000	Max. :23.000	Max. :92.00	Max. :13.000	Max. :3.0000	Max. :282.00

pm10	o3	no2	so2
Min. : 9.00	Min. : 2.00	Min. : 3.00	Min. : 0.0000
1st Qu.: 32.00	1st Qu.:15.00	1st Qu.: 6.00	1st Qu.: 0.0000
Median : 39.00	Median :19.00	Median : 8.00	Median : 0.0000
Mean : 45.36	Mean :19.93	Mean :10.06	Mean : 0.3753
3rd Qu.: 53.00	3rd Qu.:24.00	3rd Qu.:12.00	3rd Qu.: 0.0000
Max. :176.00	Max. :53.00	Max. :45.00	Max. :27.0000
	NA's :5		

Appendix B:

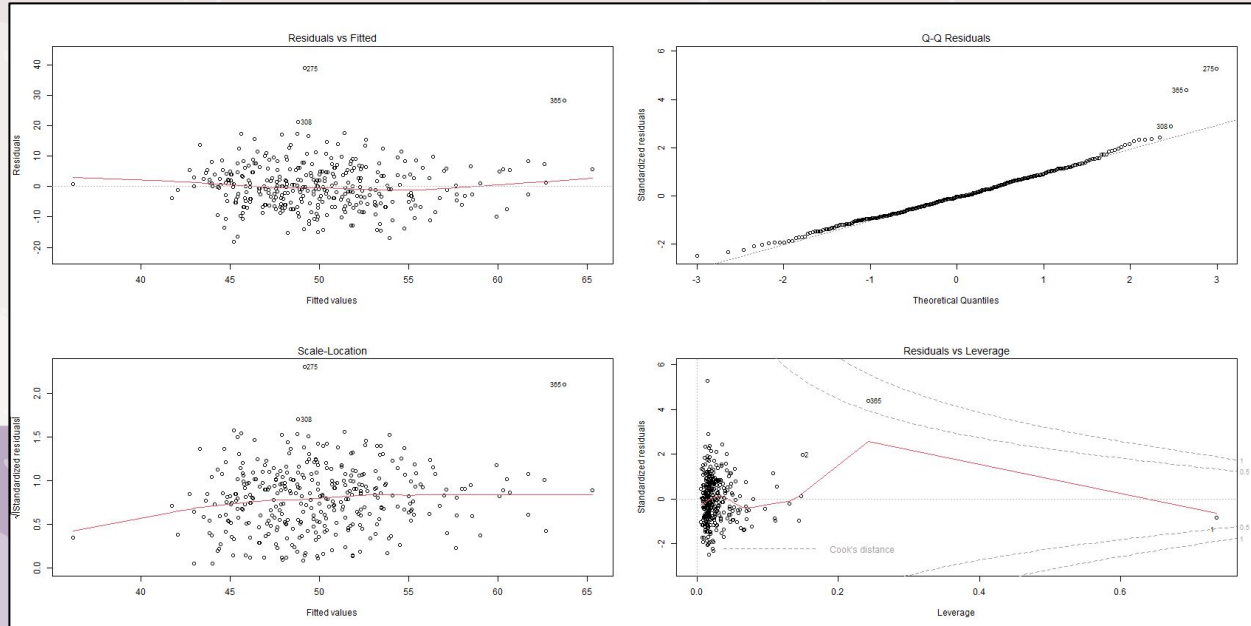
OLS Model tests and plots

```
> vif(OLSmodel)
      ard      acs      ahf      stroke      pm25      pm10      o3      no2      so2
1.080920 1.069871 1.024472 1.037529 2.707215 2.620607 1.863985 1.115462 1.066192
> bptest(OLSmodel)

studentized Breusch-Pagan test

data:  OLSmodel
BP = 14.389, df = 9, p-value = 0.1091

> durbinWatsonTest(OLSmodel)
lag Autocorrelation D-W Statistic p-value
1      0.07782565      1.802544      0.048
Alternative hypothesis: rho != 0
```



Appendix C: Stepwise selection

```
Call:
lm(formula = total_case ~ pm25 + pm10 + o3 + pm25.L1 + pm10.L2 +
    o3.L3 + no2.L3, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.0423	-2.6254	-0.3384	1.9098	12.3294

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.56429	0.64814	11.671	<2e-16 ***
pm25	0.01576	0.01115	1.414	0.1583
pm10	-0.02335	0.01368	-1.707	0.0887 .
o3	0.07963	0.03695	2.155	0.0318 *
pm25.L1	0.03757	0.01634	2.299	0.0221 *
pm10.L2	-0.05555	0.02472	-2.247	0.0252 *
o3.L3	-0.06137	0.03700	-1.659	0.0981 .
no2.L3	0.06206	0.03128	1.984	0.0480 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.484 on 354 degrees of freedom
Multiple R-squared: 0.09026, Adjusted R-squared: 0.07227
F-statistic: 5.017 on 7 and 354 DF, p-value: 0.00001943

```
> vif(Stepmodel)
      pm25      pm10      o3  pm25.L1  pm10.L2      o3.L3      no2.L3 
5.926679  2.509969  2.089547 12.001096  7.289564  2.044208  1.166526 
> bptest(Stepmodel)

studentized Breusch-Pagan test

data:  Stepmodel
BP = 7.959, df = 7, p-value = 0.3362

> durbinWatsonTest(Stepmodel)
lag Autocorrelation D-W Statistic p-value
1      0.061367      1.873937    0.158
Alternative hypothesis: rho != 0
```


Appendix D: Log transformed model

```
Call:
lm(formula = log_total_case ~ pm25 + pm10 + o3 + o3.L2 + log_total_case.L1,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.40484 -0.22460  0.02072  0.23478  0.87784

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.8836462   0.1212645   15.533  < 2e-16 ***
pm25         0.0029366   0.0008621    3.406  0.000734 ***
pm10        -0.0022434   0.0014141   -1.586  0.113536
o3           0.0075596   0.0037905    1.994  0.046880 *
o3.L2       -0.0060110   0.0038204   -1.573  0.116524
log_total_case.L1 0.0585815  0.0522602    1.121  0.263063
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3663 on 355 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.07954,    Adjusted R-squared:  0.06658
F-statistic: 6.136 on 5 and 355 DF,  p-value: 0.0000181

> vif(Logmodel)
           pm25           pm10           o3           o3.L2 log_total_case.L1
3.204901      2.426868      1.989570      1.998157      1.054185
> bptest(Logmodel)

studentized Breusch-Pagan test

data: Logmodel
BP = 3.0353, df = 5, p-value = 0.6945

> durbinWatsonTest(Logmodel)
lag Autocorrelation D-W Statistic p-value
1 -0.01001107 2.016729 0.964
Alternative hypothesis: rho != 0
```

Appendix E: Log transformed model tests

```
> vif(Interactmodel)
there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif
```

pm25_c	o3_c	pm10_c	no2_c	so2_c	ard_c	acs_c	ahf_c	stroke_c	log_total_case.L1
2.787112	1.967226	2.748703	1.137154	1.145712	1.139447	1.091088	1.040430	1.054449	1.104042
pm25_c:ard_c	pm25_c:acs_c	pm25_c:ahf_c	pm25_c:stroke_c	o3_c:ard_c	o3_c:acs_c	o3_c:ahf_c	o3_c:stroke_c		
2.228316	1.964619	1.795292	2.116535	2.041100	1.768355	1.655241	1.983059		

```
> bptest(Interactmodel)

studentized Breusch-Pagan test

data: Interactmodel
BP = 16.79, df = 18, p-value = 0.5376

> durbinWatsonTest(Interactmodel)
lag Autocorrelation D-W Statistic p-value
1 0.03860428 1.921373 0.388
Alternative hypothesis: rho != 0
```

Appendix F: Addressing curvilinear relationships - Polynomial Model

```
Call:
lm(formula = log_total_case ~ pm25_c + I(pm25_c^2) + o3_c + I(o3_c^2) +
    pm10_c + I(pm10_c^2) + no2_c + so2_c + ard_c + acs_c + ahf_c +
    stroke_c + pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c +
    o3_c:ard_c + log_total_case.L1, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.59159	-0.02948	0.02752	0.05216	0.11700

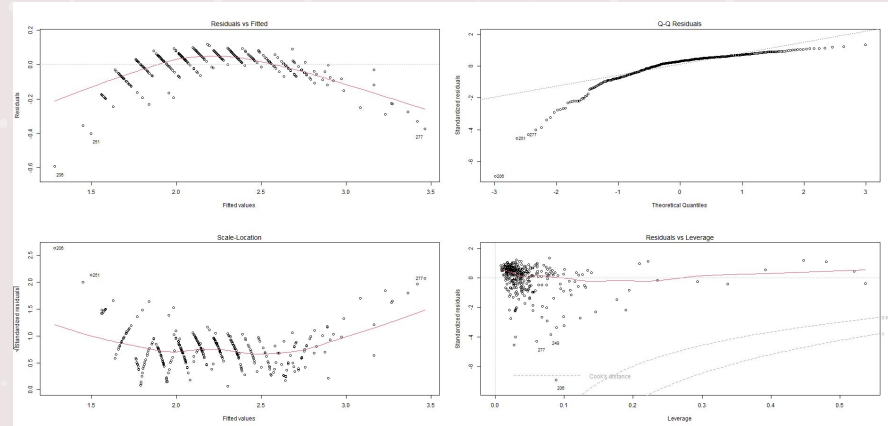
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.252499106	0.030723988	73.314	< 2e-16
pm25_c	0.000019964	0.000284507	0.070	0.944099
I(pm25_c^2)	0.000003389	0.000002326	1.457	0.146087
o3_c	-0.000900207	0.000998213	-0.902	0.367786
I(o3_c^2)	0.000030601	0.000058753	0.521	0.602813
pm10_c	0.001008580	0.000536731	1.879	0.061076
I(pm10_c^2)	-0.000013961	0.000005973	-2.337	0.020006
no2_c	-0.000274781	0.000790236	-0.348	0.728264
so2_c	0.004900451	0.008284242	0.592	0.554549
ard_c	0.101581932	0.001859393	54.632	< 2e-16
acs_c	0.099413995	0.004073302	24.406	< 2e-16
ahf_c	0.100173332	0.004660192	21.496	< 2e-16
stroke_c	0.109094583	0.003363537	32.434	< 2e-16
log_total_case.L1	-0.018521645	0.013288231	-1.394	0.164269
pm25_c:ard_c	-0.000173363	0.000063321	-2.738	0.006507
pm25_c:ahf_c	-0.000454513	0.000119599	-3.800	0.000171
pm25_c:stroke_c	-0.000341859	0.000083702	-4.084	0.0000551
o3_c:ard_c	-0.001273512	0.000354382	-3.594	0.000374

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08943 on 343 degrees of freedom
(1 observation deleted due to missingness)

Multiple R-squared: 0.947, Adjusted R-squared: 0.944
F-statistic: 360.4 on 17 and 343 DF, p-value: < 2.2e-16



Appendix G: Addressing curvilinear relationships - Splines Model

```
Call:
lm(formula = log_total_case ~ ns(pm25_c, df = 3) + ns(o3_c, df = 3) +
    ns(pm10_c, df = 3) + no2_c + so2_c + ard_c + acs_c + ahf_c +
    stroke_c + pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c +
    o3_c:ard_c + log_total_case.L1, data = data)
```

Residuals:

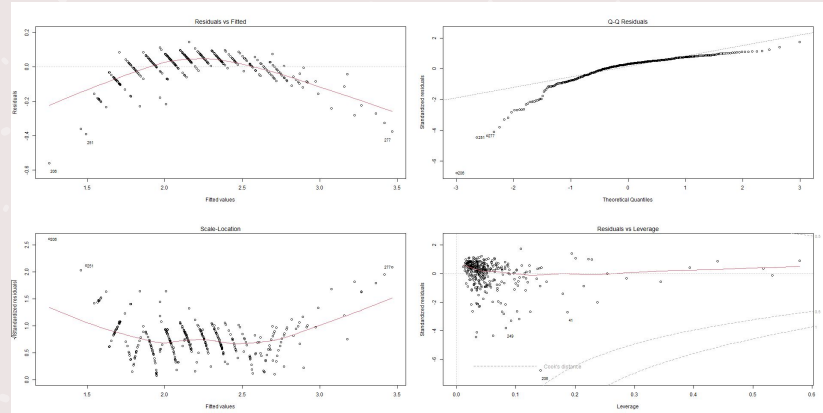
	Min	1Q	Median	3Q	Max
	-0.56033	-0.02835	0.02702	0.05207	0.14684

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.18160667	0.06565864	33.226	< 2e-16 ***
ns(pm25_c, df = 3)1	0.00998535	0.03908287	0.255	0.798498
ns(pm25_c, df = 3)2	0.02129439	0.06147927	0.346	0.729281
ns(pm25_c, df = 3)3	0.12148345	0.06431115	1.889	0.059743 .
ns(o3_c, df = 3)1	-0.04700486	0.03219530	-1.460	0.145216
ns(o3_c, df = 3)2	0.11565646	0.09169389	1.261	0.208054
ns(o3_c, df = 3)3	0.06480791	0.06551472	0.989	0.323263
ns(pm10_c, df = 3)1	0.08584783	0.04617130	1.859	0.063844 .
ns(pm10_c, df = 3)2	0.05955359	0.11598435	0.513	0.607961
ns(pm10_c, df = 3)3	-0.06959972	0.07052389	-0.987	0.324396
no2_c	-0.00028435	0.00080984	-0.351	0.725717
so2_c	0.00577019	0.00834209	0.692	0.489600
ard_c	0.10138196	0.00186403	54.389	< 2e-16 ***
acs_c	0.10007650	0.00407534	24.557	< 2e-16 ***
ahf_c	0.10081600	0.00465091	21.677	< 2e-16 ***
stroke_c	0.10947465	0.00336025	32.579	< 2e-16 ***
log_total_case.L1	-0.01901670	0.01331441	-1.428	0.154129
ard_c:pm25_c	-0.00019111	0.00006431	-2.972	0.003174 **
ahf_c:pm25_c	-0.00042372	0.00011890	-3.564	0.000418 **
stroke_c:pm25_c	-0.00034605	0.00008448	-4.096	0.0000526 ***
ard_c:o3_c	-0.00108395	0.00036443	-2.974	0.003146 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08915 on 340 degrees of freedom
 (1 observation deleted due to missingness)
 Multiple R-squared: 0.9478, Adjusted R-squared: 0.9447
 F-statistic: 308.5 on 20 and 340 DF, p-value: < 2.2e-16



Appendix H: Addressing curvilinear relationships - ANOVA Test

```
> anova(Finalmodel, Polymodel, Splinesmodel)
Analysis of Variance Table
```

```
Model 1: log_total_case ~ pm25_c + o3_c + pm10_c + no2_c + so2_c + ard_c +
  acs_c + ahf_c + stroke_c + pm25_c:ard_c + pm25_c:ahf_c +
  pm25_c:stroke_c + o3_c:ard_c + log_total_case.L1
Model 2: log_total_case ~ pm25_c + I(pm25_c^2) + o3_c + I(o3_c^2) + pm10_c +
  I(pm10_c^2) + no2_c + so2_c + ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1
Model 3: log_total_case ~ ns(pm25_c, df = 3) + ns(o3_c, df = 3) + ns(pm10_c,
  df = 3) + no2_c + so2_c + ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	346	2.7883				
2	343	2.7431	3	0.045218	1.8963	0.1299
3	340	2.7024	3	0.040681	1.7060	0.1655

- Polynomial and Spline models do not provide statistically significant improvements on the Final model